Yalu Cai

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Education

Rutgers University — Piscataway, NJ, USA
PhD in Computer Science, advised by Prof. Santosh Nagarakatte — *Sept. 2025 – Present*

Duke University — Durham, NC, USA
Master of Science in Computer Science — *Aug. 2020 – May 2022*

- GPA: 3.837/4.0

Cornell University — Ithaca, NY, USA
Bachelor of Science in Biological Engineering — *Aug. 2018 – May 2020*

- GPA: 3.898/4.3

Zhejiang University — Hangzhou, China
Bachelor of Engineering in Biological Engineering — *Sept. 2016 – Jul. 2018*

- GPA: 3.77/4.0

Publications

Proof-Carrying-BPF Programs for the eBPF Verifier
Yalu Cai, Santosh Nagarakatte
eBPF '26: The 4th Workshop on eBPF and Kernel Extensions, Prague, Czech Republic, Sept. 2026

Research

Alivio: Proof-Carrying BPF Programs | *Rust, CVC5* — Sept. 2025 – Present

- A userspace mirror of the Linux kernel's eBPF verifier that generates machine-checkable SMT proofs for the program points the kernel's abstract domains cannot prove safe, widening the class of accepted programs without growing the kernel's trusted computing base.
- Matches the kernel verifier's verdict on all 3,267 BPF selftests; its proof bundles let a patched kernel load 189 production programs from Cilium, Calico, and Inspektor Gadget that the stock verifier rejects, with $3.4\times$ faster loads than prior work.

Verified Translator from EVM Bytecode to eWASM | *Coq* — Sept. 2021 – May 2022

- Implements a translator that translates a subset of Ethereum Virtual Machine Bytecode program into WebAssembly.
- Verifies the translator for Semantic Preservation using Coq.

Typechecker for ssm-edsl | *Haskell* — May. 2021 – Oct. 2021

- Worked with the ssm-edsl group from Columbia University and Chalmers University (Sweden) to implement a typechecker for SSM (Sparse Synchronous Model), a language designed to aid real-time data collection.

Classifying Memory Access Patterns for Prefetching | *C++* — May 2021 – Sept. 2021

- Worked with Professor Calvin Lin (UT Austin) to recreate the compiler-aided prefetcher described in this ASPLOS 2020 paper.
- Implemented a program that reads an instruction trace and a ChampSim cache-miss profile and constructs the dataflow graph (prefetch kernel) for the delinquent loads.

GKAT Interpreter | *OCaml* — Feb. 2020 – May 2020

- An interpreter for GKAT (Guarded Kleene Algebra with Tests), an algebraic language for reasoning about the control flow of uninterpreted imperative programs.
- Parses GKAT expressions into automata and decides equivalence of expressions by normalizing the automata and checking bisimilarity with a union-find algorithm.

Work Experience

Software Engineer Jun. 2022 – Aug. 2025
CertiK LLC *New York, NY, USA*

- Designed and developed compilers from Solidity to Dafny and Boogie. Especially involved in the design of the intermediate representation language Swivel, which connects Solidity to both Dafny and Boogie.
- Extended Solidity with a specification language which is adapted from Java Modeling Language (JML).
- Performed deductive verification on Solidity projects translated into Dafny.
- Developed a set of Python scripts that automates the building and running of the company's formal verification tool on Solidity projects, greatly decreasing development and testing time.

Graduate Teaching Assistant Aug. 2020 – May 2022
Duke University *Durham, NC, USA*

- Teaching assistant for *CS310 Operating Systems; CS230 Discrete Math; CS334 Mathematical Foundations of CS*
- Responsible for organizing recitation sessions, holding office hours, and grading homework and exams.

Coding Projects

EVM Bytecode Compiler | *LLVM, Python* April 2024 – Now

- A compiler of EVM bytecode to machine code using LLVM (llvmlite).
- The compiler supports basic opcodes including STOP, LOAD, STORE, SUB, ADD, PUSH, POP, MSTORE, MLOAD, etc.

Tiger Compiler | *Standard ML* Jan. 2021 – May 2021

- A compiler for Tiger, a small language with nested functions, record values with implicit pointers, arrays, integer and string variables, and a few simple structured control constructs.
- Features: type checking and bounds checking for array subscriptions.

Paxos Distributed System | *Java* Jan. 2022 – May 2022

- A Java implementation of the Paxos protocol, which utilizes state machine replication to tolerate server failures in a distributed system.
- This project models the clients, servers, messages and timers in a distributed system. It also implements algorithms for stable leader election and garbage collection.

Technical Skills

Languages: Rust, C/C++, OCaml, Python, Coq, Haskell, Golang, Java, SQL, Swift, JavaScript, TypeScript
Developer Tools: Git, Linux, Docker, LLVM, SMT solvers (CVC5)